TITLE PAGE

- **Problem Statement ID –** Q1
- **Problem Statement Title-** Eco-Loops Building Agents
- **Theme-** AI-Powered Autonomous Smart Building Optimization Challenge
- **PS Category-** Software
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Eco-Loops Building Agents

Detailed Explanation of the Proposed Solution

- AI-powered Building Energy Management System integrating EnergyPlus, LangGraph, Model Context Protocol (MCP), and LLaMA 3.1.
- Uses real-time building telemetry, weather, and occupancy data to generate intelligent HVAC control decisions.

How It Addresses the Problem

- Replaces static HVAC schedules with dynamic, context-aware optimization.
- Achieves 23.3% HVAC energy savings while maintaining occupant comfort.
- Ensures safe HVAC operation through continuous monitoring and validation.

Innovation & Uniqueness

- MCP-based architecture for seamless AI and Building Management System integration.
- Smart Trigger Engine reduces unnecessary LLM calls by 53%.
- Safety Validator enforces HVAC limits, delivering 100% safety compliance.

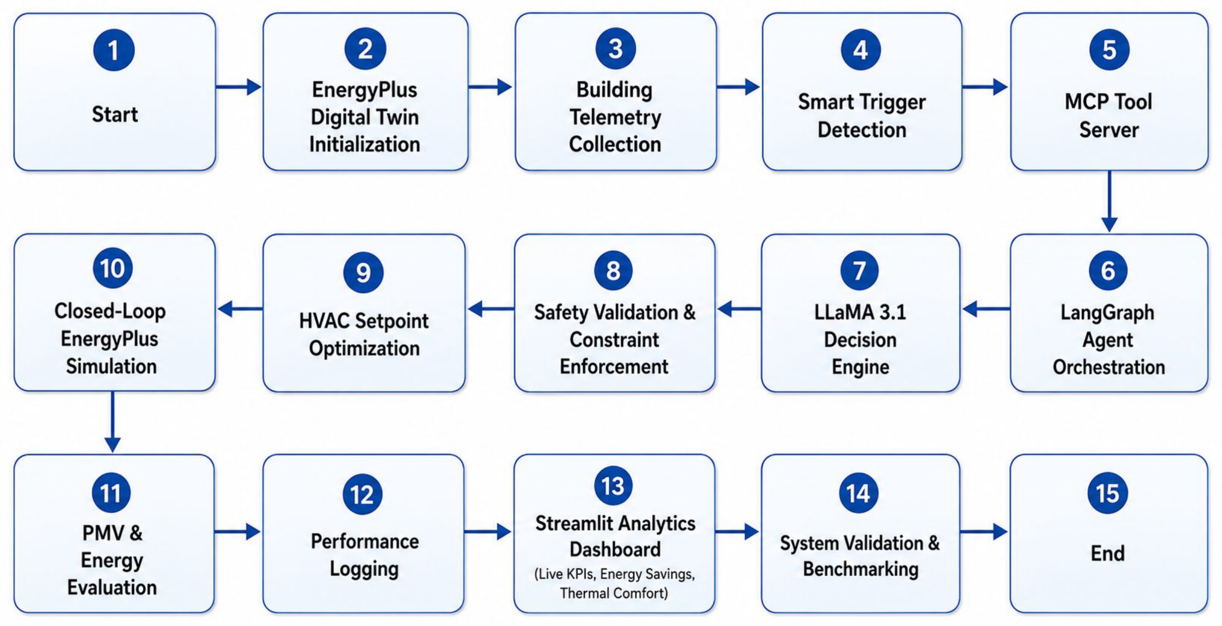
TECHNICAL APPROACH

Technologies Used

1. Simulation: EnergyPlus, PyEnergyPlus
2. AI Engine: LLaMA 3.1 (Groq), LangGraph
3. Communication: Model Context Protocol (MCP), FastMCP
4. Backend & Dashboard: Python, FastAPI, Streamlit, Plotly

Implementation Methodology

1. Collect real-time building telemetry from EnergyPlus.
2. Process data through MCP and LangGraph AI workflow.
3. Generate optimized HVAC setpoints using LLaMA 3.1.
4. Validate decisions with safety constraints.
5. Apply optimized setpoints to the building model.
6. Monitor energy savings and occupant comfort.



FEASIBILITY AND VIABILITY

Analysis of the Feasibility of the Idea

- Compatible with existing Building Management Systems (BMS)
- Uses open-source AI tools and EnergyPlus simulation
- Scalable for commercial and smart buildings

Potential Challenges and Risks

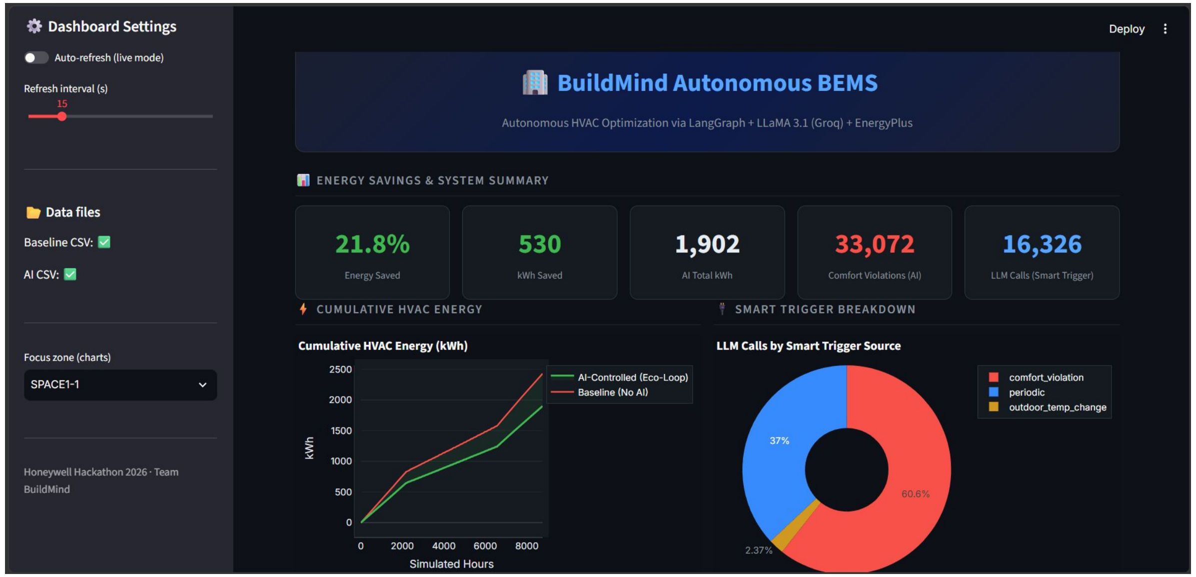
- Incorrect AI-generated HVAC control decisions
- API latency and operational costs
- Maintaining occupant comfort during optimization

Strategies for Overcoming These Challenges

- Safety validation before applying AI decisions
- Smart triggering to reduce unnecessary AI calls
- PMV-based thermal comfort monitoring

ARTIFACTS

DashBoard Screenshot



- Performance Results achieving 23.3% energy savings, 53% fewer LLM calls, and 100% safety compliance.

RESEARCH AND REFERENCES

- International Energy Agency (IEA) – Building Energy Statistics
- U.S. Department of Energy (DOE) – Building Energy Data Book
- EnergyPlus – Official Documentation
- EnergyPlus – Engineering Reference
- Model Context Protocol (MCP) – Official Specification
- FastMCP – Python MCP Framework
- Groq – LLaMA 3.1 Documentation
- ASHRAE Standard 55 – Thermal Comfort Standard
- UC Berkeley CBE – Thermal Comfort Tool
- NREL & Recent Research – AI for Building Energy Management